

Food for Thought

the relationship between SNAP benefits and poverty rate for minorities in MO

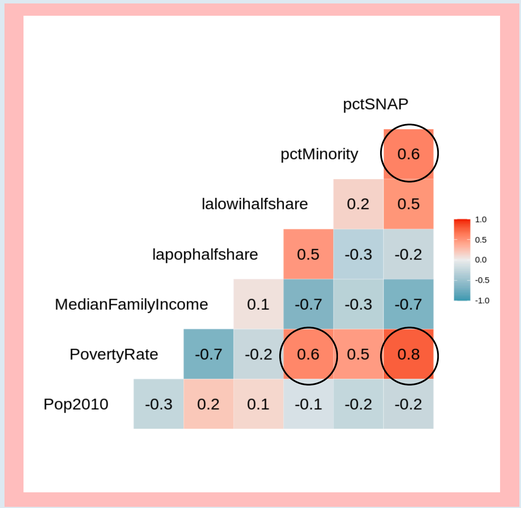
Elyse Freeman - DSA 7020



01. The Data

The data explored in this project comes from the USDA Food Access Research Atlas.

This project aims to show the relationship between percent of housing units per census tract on SNAP (pctSNAP) , Poverty Rate, and percent minority population per Census Tract (pctMinority). Secondary factors that are being considered are lapophalfshare (low-access, more than ½ from a supermarket) and lalowihalfshare (low-access and low-income). Below is the correlation matrix for the selected variables. There are strong correlations between pctSNAP and PovertyRate, pctSNAP and pctMinority, and lalowihalfshare and PovertyRate.



02. Visualizations of Key Correlations with pctSNAP

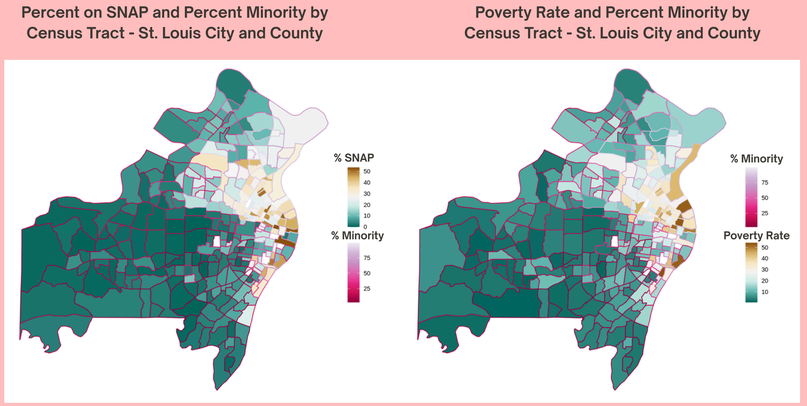


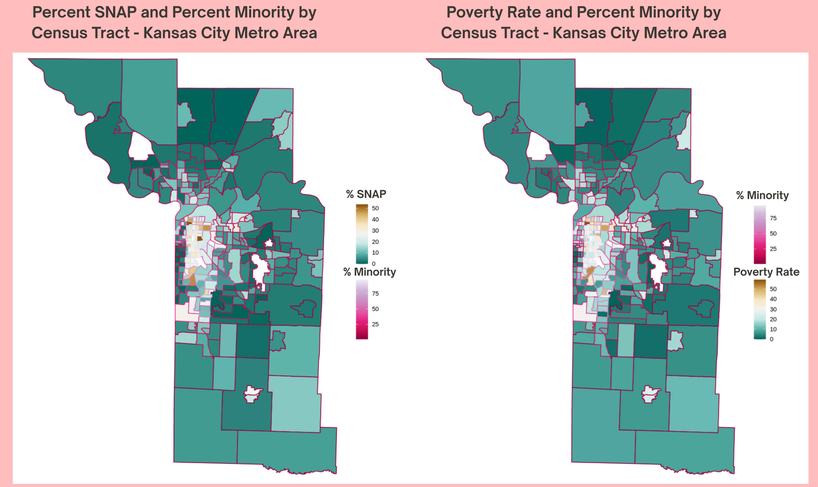
These plots show the key variables considered in the later models - pctMinority, PovertyRate, lalowihalfshare, and lapophalfshare compared to pctSNAP.

- pctSNAP and PovertyRate have a correlation of 0.7757. The points are clustered/centered around the line of best fit.
- pctMinority and PovertyRate have a correlation of 0.5946 and the points are clustered in the bottom left side/corner.
- lalowihalfshare and pctSNAP have a correlation of 0.4960. The points are clustered in the bottom of the grid, with outliers towards the top.
- lapophalfshare and pctSNAP is the only negative correlation at -0.2272. Many of the points are clustered to the bottom right side which differs from the other three plots. This implies that there are higher amounts of housing units in a census tract that are low access to a supermarket, but not on SNAP benefits.
- There are distinct clusters of urban/rural points, especially in pctMinority and lalowihalfshare.

03. A closer look at St. Louis and Kansas City Metro Areas

The plots show that areas with higher percent SNAP also have a higher Poverty Rate. Additionally, areas with higher percentages of minorities have higher percent SNAP and Poverty Rates.





04. Fitting a Model For pctSNAP

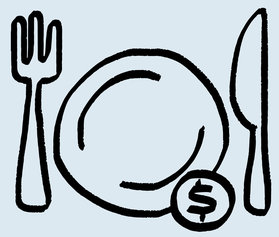
Examining fit5(summary shown) further shows a standard error of the estimate for the testing set of 2.9563 and for the training set of 4.6739. The testing error is less than the training error which could be due to using a smaller sample size. The t values are far enough from 0, especially relative to the standard error, which leads to rejecting our null hypothesis that the variables have no effect on pctSNAP . It can be concluded that fit5 does a good job at predicting pctSNAP values.

```
Call:
lm(formula = pctSNAP ~ PovertyRate * lalowihalfshare + pctMinority +
    lapophalfshare, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-33.352  -2.894   0.050   2.710  24.370

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  0.8857348  1.0627620   0.833    0.405
PovertyRate                  0.5966861  0.0399158  14.949  < 2e-16 ***
lalowihalfshare              0.2943658  0.0260718  11.291  < 2e-16 ***
pctMinority                  0.1271479  0.0081450  15.610  < 2e-16 ***
lapophalfshare              -0.0584836  0.0124917  -4.682 3.26e-06 ***
PovertyRate:lalowihalfshare -0.0062976  0.0008478  -7.428 2.44e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.602 on 953 degrees of freedom
Multiple R-squared:  0.7133,    Adjusted R-squared:  0.7118
F-statistic: 474.3 on 5 and 953 DF,  p-value: < 2.2e-16
```



05. Conclusions

The percent of people in a tract on SNAP does have a meaningful relationship with Poverty Rate and percent Minority. Using the latter variables as well as accounting for food access, a model was fit. Additionally, it can be visually shown that PovertyRate & pctMinority correlate with pctSNAP. With benefits such as SNAP in the news, it is important to know which communities will be most impacted. The plots in #3 show specific communities that are high minority with a high poverty rate and high percentage of housing units on SNAP. These are likely marginalized communities that will be impacted by changes to SNAP. Knowing where these communities are can lead to action steps like increased donations to local food pantries.